SWINBURNE UNIVERSITY OF TECHNOLOGY

Exploratory Data Analysis Report
Anti-Spoofing (Liveness) Dataset

Facial Recognition with Emotion and Liveness
COS30082 Applied Machine Learning, Group Project

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Source notebook: anti_spoofing_eda.ipynb

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Data: Google Drive folder “Anti-Spoofing” (LCC FASD: training, development, evaluation)

1. Summary

This report describes the data that our team will use to train the liveness (anti-spoofing) model. The model must decide whether a face in front of the camera is a **real** person or a **spoof**, for example a video of a person replayed on a phone or screen. Before training, we checked what is in the data: how many images each class has, whether the files are clean, whether real and spoof images differ in ways a model could cheat with, who and what is in the videos, and whether the test data was seen during training. All numbers come from running the project notebook on the real dataset.

Question	Answer	Verdict
How big is the dataset?	18,827 images: 8,299 training, 2,948 validation, 7,580 test. A validation split already exists	Large enough
Are the files clean?	No corrupted files. All scanned images are PNG, RGB. Sizes vary a lot (673 different sizes)	Good, needs resizing
Is it balanced?	No. Spoof is 85% of training and 96% of the test set. Only 314 real test images	Imbalanced
Is image quality good?	About 19% of scanned images are blurry (mostly spoofs); very few are too dark or too bright	Acceptable
Can a simple number separate real from spoof?	No single measurement passes the risk level (strongest: contrast, 0.67 of 1.0)	Low shortcut risk
Is there data leakage?	9 people appear in both training and validation; 4 near-identical images are shared. Test is clean on the images we could compare, but its file names carry no person information	Needs action

Main message
The data is usable, and no obvious shortcut separates real from spoof. The main risks are (1) a **very unbalanced test set** (96% spoof), so accuracy alone would look excellent for a model that always says “spoof”, (2) **people shared between training and validation**, which makes the validation score optimistic, and (3) a **test set with anonymised file names**, so we cannot check people or devices there.

Important limit of this run
The quality scan opened a random **10%** of the images (1,883 of 18,827), because these images are large and Google Drive is slow. Percentages are reliable estimates; counts of duplicates are **minimum** values. The people and video counts use file names only and cover all images. A full scan (using a zip file) will give exact numbers.

2. What data we have and how we checked it

The shared folder follows the LCC FASD (Large Crowd-sourced Face Anti-Spoofing Dataset) layout. It has three folders, *training*, *development* and *evaluation*, and each contains a *real* and a *spoof* sub-folder. We use them as the training, validation and test splits. Real images show a live person; spoof images are frames from videos in which a person is shown to the camera on another device. In the training and development folders

the file name tells us the person, the recording session, the frame number and the devices used for the attack. In the evaluation folder the names are anonymised (*real_41.png*, *spoof_800.png*).

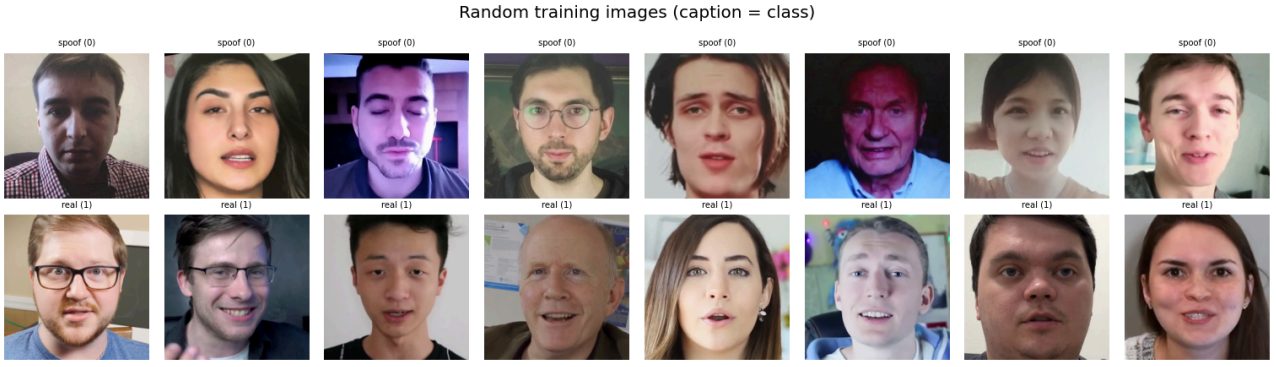


Figure 1. Random training images. Top row: spoof. Bottom row: real. The two classes look very similar to the human eye.

How the checks work

Check	Plain-language meaning	How it is measured
Size / format	Are all images the same shape and file type?	Width, height, file type and colour mode
Sharpness	Is the picture blurry?	Variance of the Laplacian (an edge detector) after resizing to 224 x 224. Low = blurry
Brightness / contrast / colour	Too dark, too bright, washed out, or too grey?	Average and spread of grey levels (0 to 255); colour saturation
Shortcut check	Can one simple number tell real from spoof?	AUC of each measurement: 0.5 = no information, 1.0 = perfect separation
People and videos	How many different people and videos are behind the images?	Read from the file names
Duplicates	Is the same picture stored twice?	MD5 hash = identical file; perceptual hash = near copy
Leakage	Do test images, or the same people, appear in training?	Compare hashes and person numbers between splits

3. Class distribution

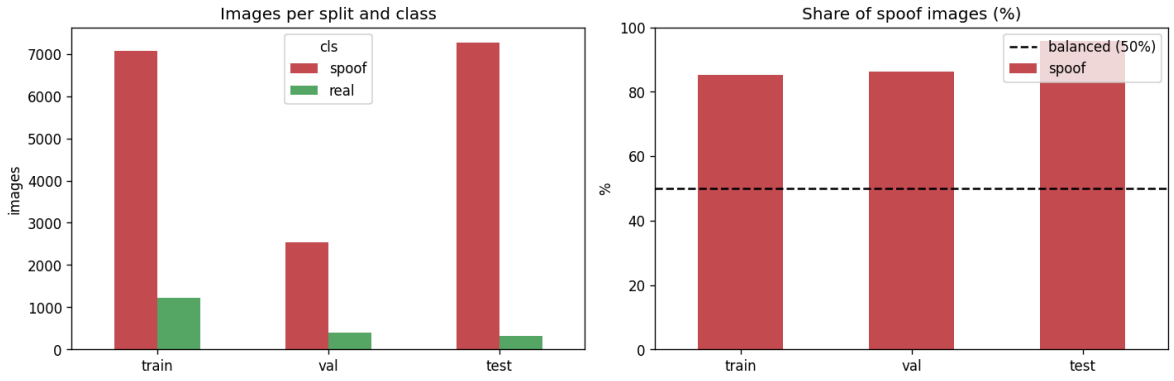


Figure 2. Images per split and class (left) and share of spoof images per split (right). The dashed line marks a balanced 50%.

Split	Real	Spoof	Total	Spoof share
Training	1,223	7,076	8,299	85.3%
Validation	405	2,543	2,948	86.3%
Test	314	7,266	7,580	95.9%
Total	**1,942**	**16,885**	**18,827**	**89.7%**

Spoof images are by far the larger class in every split. In training the imbalance ratio is 5.8 (7,076 spoof against 1,223 real). The test set is even more extreme: only 4.1% of its images are real, which is just 314 images. The suggested class weights are 0.59 for spoof and 3.39 for real.

> **What this means for training and testing**
> A model that answers “spoof” for every image would be right 96% of the time on the test set. Accuracy is therefore not a useful score here. We should report **APCER** (spoofs accepted as real), **BPCER** (real faces rejected), their average **ACER**, and AUC / EER, and use class weights or balanced sampling during training. Results for the real class rest on only 314 test images, so they carry more uncertainty.

4. Image properties and quality

4.1 Size, format and quality flags

All 1,883 scanned images are PNG files in RGB colour, but their sizes are very different: there are 673 distinct sizes, and the most common one (400 x 400) covers only 19 images. The typical image is about 500 x 500 pixels. All images must therefore be resized and cropped to one input size before training. This is the opposite of the emotion data, where every image has the same size.

Flag	Train	Val	Test	Meaning
Corrupted	0	0	0	File cannot be opened
Low resolution (<128 px)	0	0	1	Image too small
Blurry (threshold 50)	114	55	190	Share of scanned images: 14%, 17%, 25%
Too dark (mean <40)	2	1	6	Very dark images
Too bright (mean >210)	0	2	0	Over-exposed images
Low contrast (std <20)	0	0	0	Flat, washed-out images
Images scanned	805	323	755	10% random sample

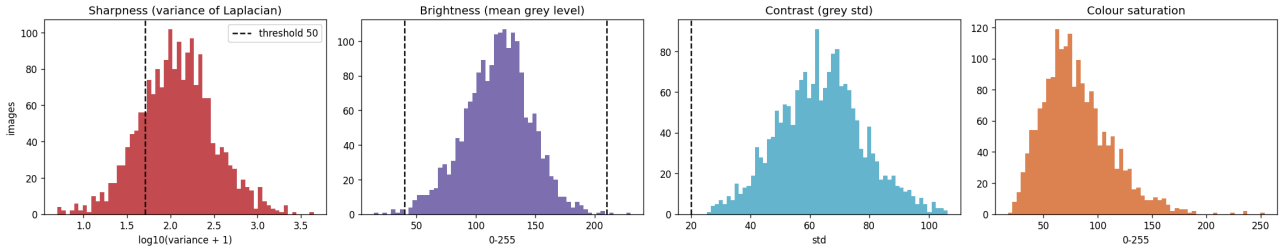


Figure 3. Distribution of sharpness, brightness, contrast and colour saturation. Dashed lines are the thresholds used to raise a flag.

Brightness, contrast and colour saturation are smooth, single-peaked curves with only small tails, so lighting is generally good. Sharpness is spread widely, and the blur threshold of 50 sits inside the left part of the curve, so it flags about 19% of the scanned images (359 of 1,883).

4.2 Looking at the extreme images

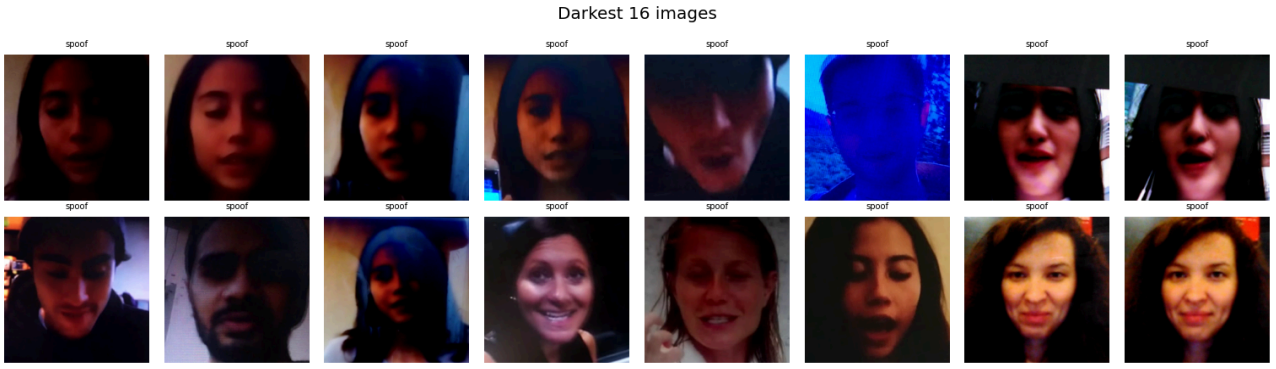


Figure 4. The 16 darkest images (all spoof). Faces are dark but still visible; some have a strong colour cast.

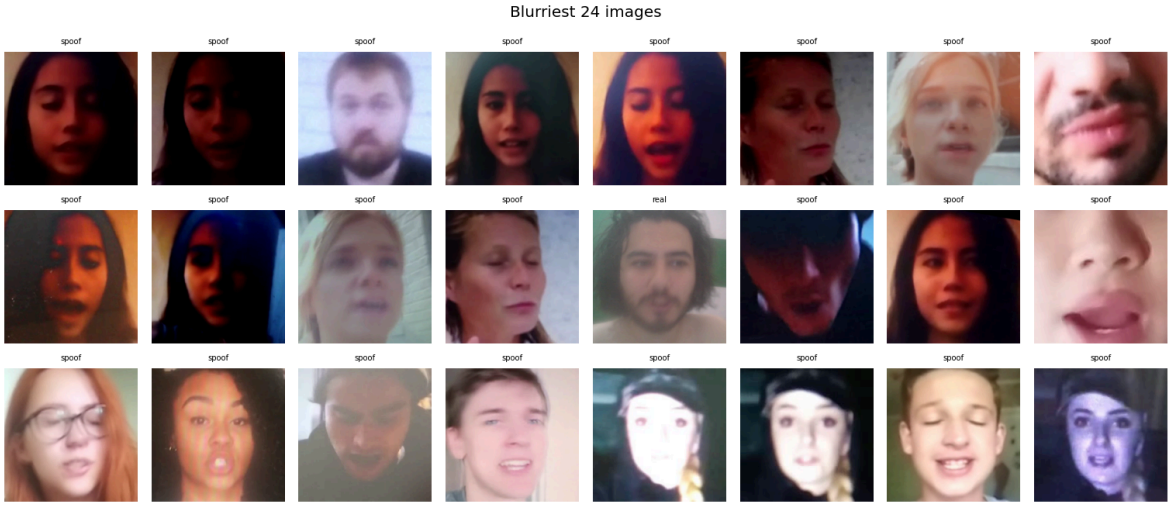


Figure 5. The 24 blurriest images (23 spoof and 1 real). They are soft, but the faces are still recognisable.

> **Finding: blur is part of the attack, so do not delete it**
> Of the blurry images, 350 are spoof (21% of scanned spoofs) and only 9 are real (5% of scanned real images). Replaying a video on a screen and filming it again makes the picture softer, so blur is a normal sign of a spoof, not a data error. The test set has the highest share of blurry images (25%), partly because it contains the most spoofs. We recommend keeping these images and using blur and compression augmentation, so that the model learns more than “soft means spoof”.

5. Real versus spoof

5.1 Can a simple number give the answer away?

In anti-spoofing, a model can reach a high score by learning a trick instead of real liveness cues, for example “large images are real” or “small files are spoofs”. We tested this by measuring how well each simple number separates the two classes on its own (AUC, where 0.5 means no information and 1.0 means perfect). The table shows the result.

Measurement	Median real	Median spoof	Separability	Risk
Contrast	56.6	64.5	0.672	low
Colour saturation	78.9	75.4	0.655	low
Sharpness	179.0	110.9	0.642	low
Brightness	109.3	120.8	0.627	low
File bytes per pixel	0.97	0.96	0.562	low
Width / height	523	488	0.559	low

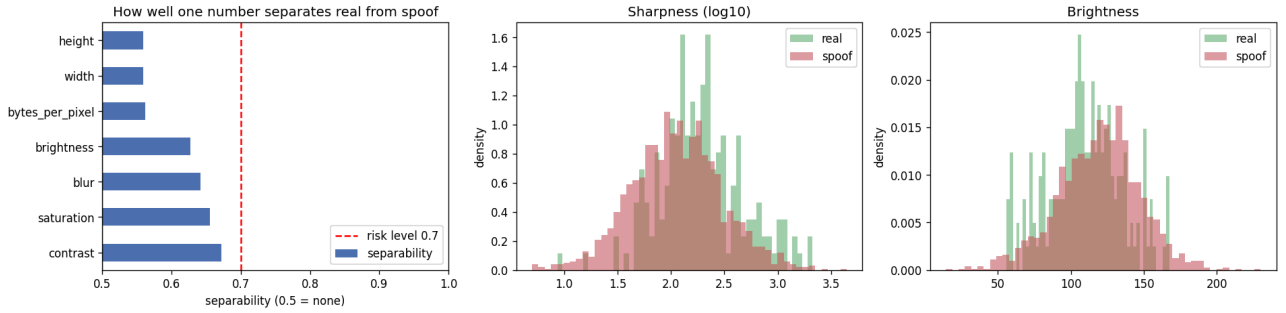


Figure 6. How well one number separates real from spoof (left) and the sharpness and brightness curves of the two classes (middle, right). The curves overlap strongly.

No measurement reaches the risk level of 0.70. Contrast is the strongest at 0.67, followed by colour saturation, sharpness and brightness. Spoof images are somewhat less sharp, a little brighter and have more contrast, which fits the picture of a recaptured screen, but the curves overlap so much that the differences cannot be used alone. Image size and file size, which are the usual shortcuts, are almost useless here (0.56). This is good news: the model will have to learn real visual cues.

5.2 What the two classes look like

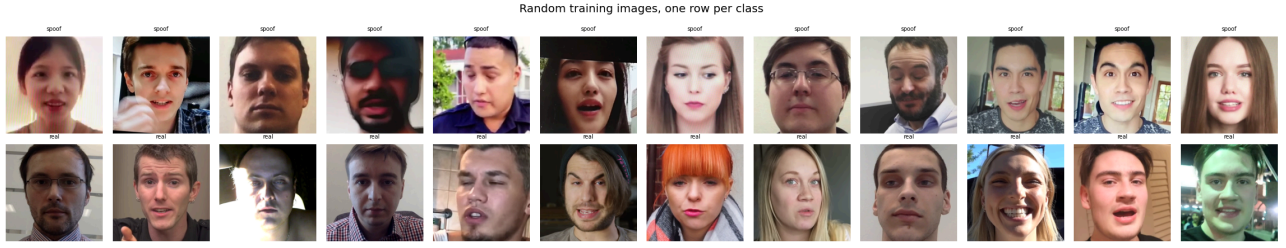


Figure 7. Random training images, first row spoof and second row real.



Figure 8. The average image of each class. The two averages are almost identical.

The average images of the two classes are both centred blurry faces and are hard to tell apart. Faces are in a similar position in both classes, which is good: position cannot be used as a hint. Because the difference between real and spoof is so subtle, the model will need enough capacity and good augmentation, and a high score should be checked on a camera or attack device it has not seen.

6. People, videos and attack devices

Split	Class	Images	People	Videos	Images per video
Training	Real	1,223	110	133	9.2
Training	Spoof	7,076	104	859	8.2
Validation	Real	405	27	42	9.6
Validation	Spoof	2,543	9	300	8.5
Test	Real	314	not available	not available	-
Test	Spoof	7,266	not available	not available	-

Each video contributes about 8 to 10 images, and consecutive frames of one video are almost the same picture. The 18,827 images therefore hold far fewer independent samples than the image count suggests. In training, 92 people have both real and spoof videos, so the model can compare the live and the recaptured version of the *same* person, which is helpful. In validation, the 2,543 spoof images come from only **9 people**, so the validation score for spoofs is based on very few identities.

> **The test set has anonymised file names**
> Test files are called *real_41.png* and *spoof_800.png*. They contain no person, session or device information, so we cannot count people or videos in the test set, check whether it shares people with training, or run a cross-device test. (The “1 video” that appears in some notebook tables for the test set is only the result of grouping all unnamed files together.)

The training and validation names show many different recording devices, but one source dominates: YOUTUBE accounts for 6,724 of 8,299 training images and 1,798 of 2,948 validation images, and FT720P adds most of the rest. A model trained on this data has seen mostly web-video material, so results on other cameras should be tested separately. In addition, 108 different attack-device combinations were found in the names; because the first part of some names is a device code instead of a source, the source column in the notebook mixes both and should be read as a rough guide only.

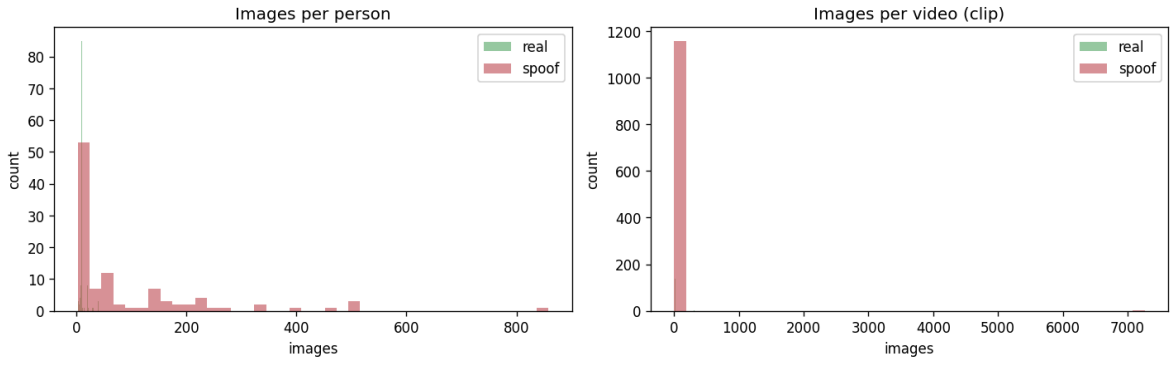


Figure 9. Images per person (left) and per video (right). A few people supply hundreds of frames. The single tall bar in the right chart is the group of unnamed test files.

7. Duplicates and data leakage

Duplicate groups	Exact (MD5)	Near (pHash)
Same video (neighbouring frames)	0	18
Same class, different videos	0	11
Across splits	0	2
Real and spoof (same picture, both labels)	0	3
Total	**0**	**34**

No two files are identical (0 exact duplicates), so the images were saved from different frames. The 34 near-duplicate groups are mostly neighbouring frames of one video, which is expected.

7.1 Same picture as real and as spoof



Figure 10. Near-identical pictures filed as spoof and as real (the label is the caption). Each pair shows the same face and pose.

Three groups contain a picture that appears both as spoof and as real. In this dataset, spoofs are recordings of replayed real videos, so a real frame and its replayed copy can look almost the same. These pairs are probably *not* labelling mistakes, but they show how small the visual difference between the classes can be. The team should confirm by eye that these labels are right.

7.2 Leakage between splits

Compared splits	Identical images	Near copies	People in both	Verdict
train vs val	0	4	9	**Leakage**
train vs test	0	0	0 (not checkable)	No image leakage found
val vs test	0	0	0 (not checkable)	No image leakage found

Training and validation share **9 people** and 4 near-identical images. The model has therefore already seen some of the validation people during training, and the validation score will look somewhat better than a score on new people. The test set shares no identical or near-identical images with the other splits in our 10% sample; for people we cannot check, because the test names carry no person information.

> **Why this matters**
> A liveness system must work for people it has never seen. If validation contains training people, the validation score is too optimistic and the decision threshold chosen on it will be too confident. Because of the sample, the true number of shared images is higher than 4.

8. Recommendations

#	Action	Priority	Who decides
1	Do not use accuracy alone. Report APCER, BPCER, ACER, AUC and EER; choose the decision threshold on validation only	High	All
2	Use class weights from class_weights.csv or balanced sampling (ratio 5.8)	High	Model team
3	Resize and crop every image to one input size (673 sizes found) and encode both classes the same way	High	Model team
4	Move the 9 people shared by train and validation out of one of the two splits, or accept an optimistic validation score and say so	High	Team agreement
5	Run the notebook on a zip copy (SCAN_FRACTION = 1.0) to get exact duplicate and leakage counts	High	Data team
6	Keep blurry and dark images; add blur, brightness, colour-jitter and JPEG-compression augmentation	Medium	Model team
7	Cap the number of frames taken from one video, since neighbouring frames are almost identical	Medium	Model team
8	Ask the data owner for the test-set names or a mapping to find people and devices; if not possible, state this limit in the report	Medium	Team lead
9	Check by eye the 3 real / spoof near-identical groups and the 4 train / validation near-copies	Low	Data team

9. Limitations

(1) Only 10% of images were scanned, so duplicate and leakage counts are lower bounds and the quality percentages for the real class rest on 181 images (27 in the test split). (2) Test file names are anonymised, so people, videos and devices could not be counted for the test set. (3) The person identifier is built from the file name (source plus person number); if numbers are not unique across sources the people count may be too high. (4) The "source" field mixes real sources and device codes. (5) Quality thresholds are starting values chosen by us. (6) Age, gender and ethnicity are not labelled and were not inferred.

Glossary

Term	Simple explanation
Real / live face	A person standing in front of the camera
Spoof (presentation attack)	A fake shown to the camera, such as a photo or a video on a screen
APCER	Share of spoofs wrongly accepted as real
BPCER	Share of real faces wrongly rejected
ACER	Average of APCER and BPCER
EER	Error rate at the threshold where APCER equals BPCER
AUC	How well a score separates two classes: 0.5 = chance, 1.0 = perfect
Shortcut	An easy trick (like image size) that a model can learn instead of the real task
Hash (MD5)	Short fingerprint of a file; identical files have identical fingerprints
Perceptual hash	Fingerprint of how an image looks; survives resizing and re-saving
Laplacian variance	Sharpness score; low means blurry
Data leakage	Test or validation data (or people) that the model has already seen during training

Generated from the project notebook outputs (EDA_report.html, EDA_summary.md and CSV tables in the team Drive folder).